COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	07/17/26	**DATE SUBMITTED:**	07/17/26
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ACTIVITY NO. 2.1 ARRAYS, POINTERS AND DYNAMIC MEMORY ALLOCATION

1. PROCEDURE OUTPUT

What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of concepts, grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I only as the AI Assistant about some concepts on how a specific topic works, also ask assistance for me to explain the observation by making sure that my grammar is correct**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Procedure

Discuss what is done by loop A and loop B in table 2-3. Additionally, discuss the output and whether the functions are working as intended. If any corrections were made, further provide your modification and analysis in table 2-4.

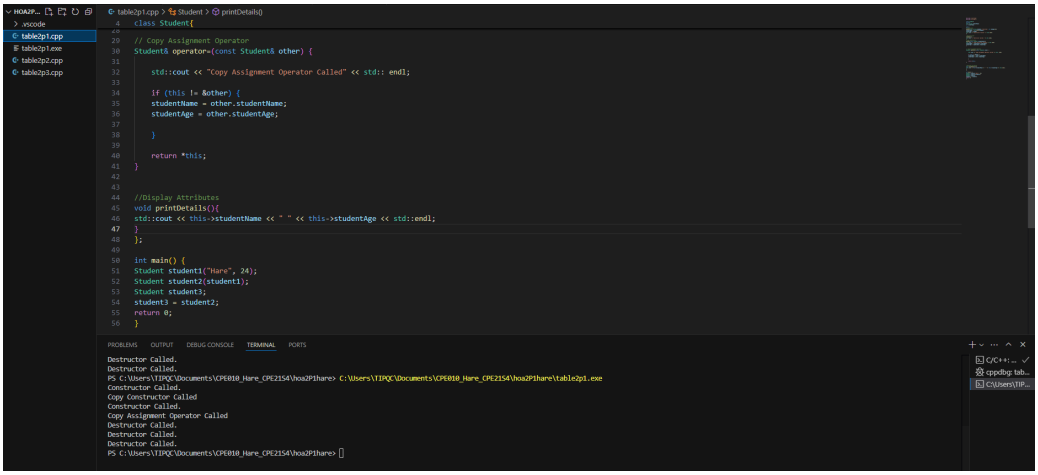
Screenshot	
Observation	The program ran without errors based on the code given. It first called the constructor for student1, then the copy constructor for student2, and the constructor again for student3. Then after that, the copy assignment operator was called when student3 = student2. At the end of the program, the destructors were called for all three objects.

Table 2-1. Initial Driver Program

Screenshot

table2p2.cpp > main()

```cpp
#include <iostream>
#include <string.h>

class Student{
private:
std::string studentName;
int studentAge;

public:
//constructor
Student(std::string newName ="John Doe", int newAge=18){
studentName = std::move(newName);
studentAge = newAge;
std::cout << "Constructor Called." << std::endl;
};

//deconstructor
~Student(){
std::cout << "Destructor Called." << std::endl;
}
//Copy Constructor
Student(const Student &copyStudent){
std::cout << "Copy Constructor Called" << std::endl;
studentName = copyStudent.studentName;
studentAge = copyStudent.studentAge;
}
//Display Attributes
void printDetails(){
std::cout << this->studentName << " " << this->studentAge << std::endl;
}
};


int main() {
const size_t j = 5;
Student studentList[j] = {};
std::string namesList[j] = {"Garcia", "Hare", "Ines", "Lacsamana", "Larracas"};
int ageList[j] = {15, 16, 18, 19, 16};

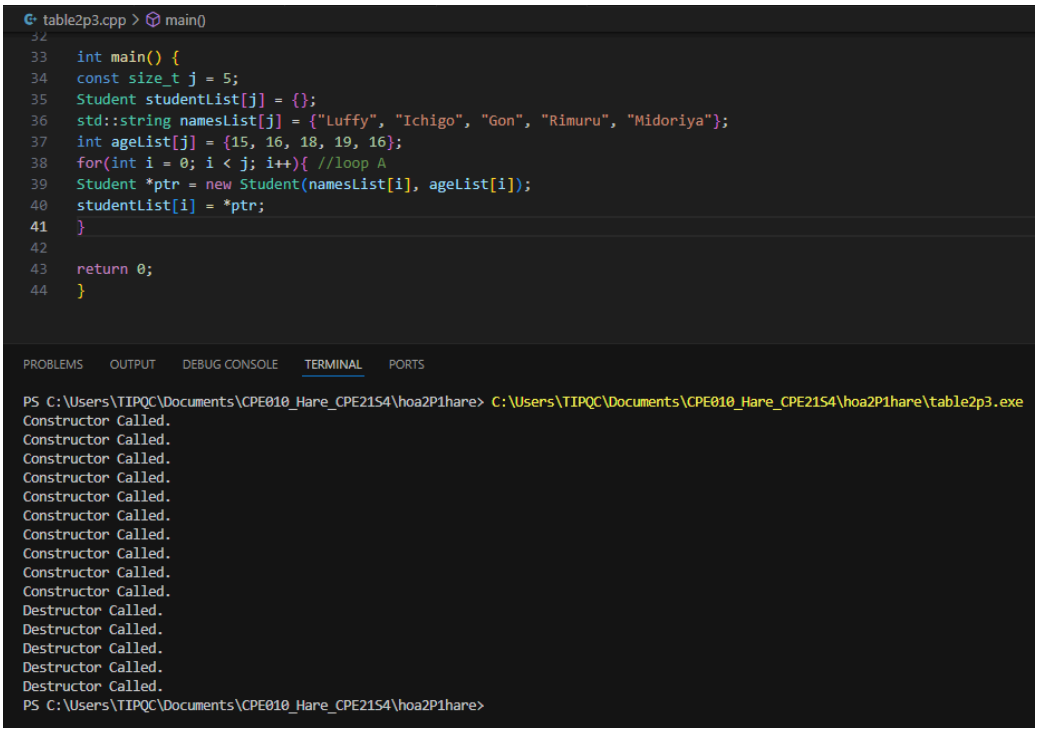
return 0;
}

```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\TIPQC\Documents\CPE010_Hare_CPE21S4\hoa2P1hare> C:\Users\TIPQC\Documents\CPE010_Hare_CPE21S4\hoa2P1hare\table2p2.exe
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
PS C:\Users\TIPQC\Documents\CPE010_Hare_CPE21S4\hoa2P1hare>
```

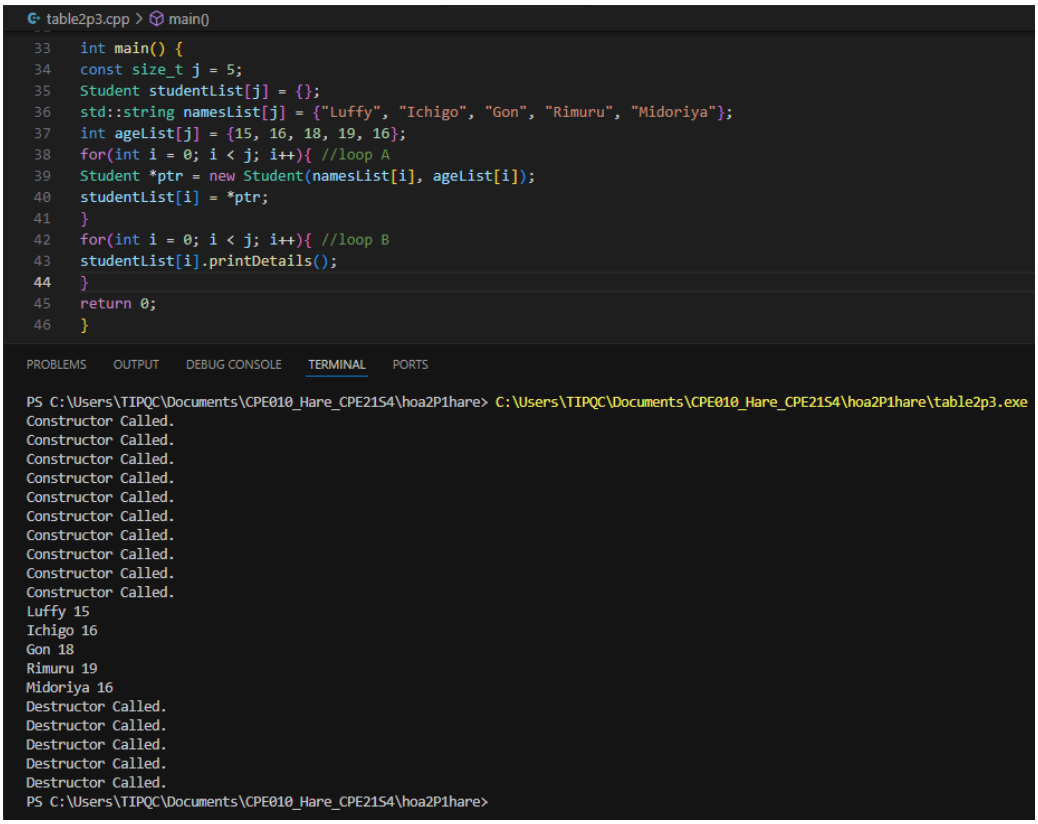
Observation	I think the program only created the student objects, so it showed the constructor and destructor messages. The names and ages were not displayed because they were not used yet.

Table 2-2 Modified Driver Program with Student Lists

Loop A	
Observation	Loop A creates a new student using the names and ages from the arrays, then saves each one into studentList. I noticed that the constructor message is printed every time a new student is created.

Loop B	table2p3.cpp > main()
	33 int main() {
	34 const size_t j = 5;
	35 Student studentList[j] = {};
	36 std::string namesList[j] = {"Luffy", "Ichigo", "Gon", "Rimuru", "Midoriya"};
	37 int ageList[j] = {15, 16, 18, 19, 16};
	38 for(int i = 0; i < j; i++){ //loop B
	39 studentList[i].printDetails();
	40 }
	41
	42 return 0;
	43 }
	
	PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
	PS C:\Users\TIPQC\Documents\CPE010_Hare_CPE21S4\hoa2P1hare> C:\Users\TIPQC\Documents\CPE010_Hare_CPE21S4\hoa2P1hare\table2p3.exe
	Constructor Called.
	Constructor Called.
	Constructor Called.
	Constructor Called.
	Constructor Called.
	John Doe 18
	John Doe 18
	John Doe 18
	John Doe 18
	John Doe 18
	Destructor Called.
	Destructor Called.
	Destructor Called.
	Destructor Called.
	Destructor Called.
	PS C:\Users\TIPQC\Documents\CPE010_Hare_CPE21S4\hoa2P1hare>

Observation	Loop B prints all the student information stored in the array. The output shows each student's name and age one by one.

Output	
Observation	For me, the program worked correctly after putting both loops inside main(). It first created the student objects, then displayed all five students with their corresponding ages.

. Table 2-3. Final Driver Program

Screenshot	
Observation	For me I didn't make any changes or modifications because the program was already working correctly.

Table 2-4 Modifications/Corrections Necessary

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Supplementary Activity

ILO C: Solve programming problems using dynamic memory allocation, arrays and pointers

Jenna's Grocery List

Apple PHP 10 x7
Banana PHP 10 x8
Broccoli PHP 60 x12
Lettuce PHP 50 x10

Jenna wants to buy the following fruits and vegetables for her daily consumption. However, she needs to distinguish between fruit and vegetable, as well as calculate the sum of prices that she has to pay in total.

Problem 1: Create a class for the fruit and the vegetable classes. Each class must have a constructor, deconstructor, copy constructor and copy assignment operator. They must also have all relevant attributes (such as name, price and quantity) and functions (such as calculate sum) as presented in the problem description above.

Problem 2: Create an array GroceryList in the driver code that will contain all items in Jenna's Grocery List. You must then access each saved instance and display all details about the items.

Problem 3: Create a function TotalSum that will calculate the sum of all objects listed in Jenna's Grocery List.

Problem 4: Delete the Lettuce from Jenna's GroceryList list and de-allocate the memory assigned.

Problem 5: Discuss the

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